Backtesting

Backtesting evaluates a trading strategy against past market data before you risk real capital. With Upstox APIs you can assemble everything a backtest needs — the correct instrument identifiers and historical OHLC candles for both equities and derivatives, including contracts that have already expired.

This guide ties together three building blocks:

Step	API	Purpose
Resolve instruments	Instrument Search	Find the `instrument_key` and `trading_symbol` for an underlying, stock, or contract.
Discover expired contracts	Get Expiries, Get Expired Option Contracts, Get Expired Future Contracts	List past expiry dates and the contracts that settled on them.
Fetch OHLC candles	Historical Candle Data V3, Get Expired Historical Candle Data	Retrieve open, high, low, close, volume, and open interest for active or expired instruments.

Upstox Plus required for expired instruments

The Expired Instruments APIs — Get Expiries, Get Expired Option/Future Contracts, and Get Expired Historical Candle Data — are available on the **Upstox Plus** plan. Instrument Search and Historical Candle Data V3 are available on all plans. See the Help Center for Plus details.

Understanding instrument identifiers

Every backtest starts by resolving the instrument you want to test. Upstox uses three identifiers:

Identifier	Example	Where it comes from
`instrument_key`	`NSE_INDEX|Nifty 50`, `NSE_EQ|INE002A01018`, `NSE_FO|53806`	Uniquely identifies an active instrument across all APIs. Returned by Instrument Search.
`trading_symbol`	`Nifty 50`, `RELIANCE`, `RELIANCE FUT 30 MAR 26`	Human-readable symbol for display and lookup. Returned alongside `instrument_key`.

Identifier	Example	Where it comes from
`expired_instrument_key`	`NSE_FO|53806|24-04-2025`	Identifies a contract that has already expired. Returned by Get Expired Option Contracts and Get Expired Future Contracts.

Use Instrument Search to convert a name or symbol into the exact `instrument_key` you need. For example, to find the underlying key for Reliance:

```
curl --location 'https://api.upstox.com/v2/instruments/search?
query=RELIANCE&exchanges=NSE&segments=EQ&records=5' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

The response includes `instrument_key` (`NSE_EQ|INE002A01018`) and `trading_symbol` (`RELIANCE`), which you carry into the next step. See the Instrument Search page for reliable query patterns and filters.

Backtest equities (cash segment)

For active equities and indices, resolve the key and pull candles directly — no expiry handling is needed.

Step 1 — Resolve the instrument key

Search for the stock and read its `instrument_key` from the response:

```
curl --location 'https://api.upstox.com/v2/instruments/search?
query=RELIANCE&exchanges=NSE&segments=EQ&records=5' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

Step 2 — Fetch historical candles

Pass the key (URL-encoded) to Historical Candle Data V3 with a `unit`, `interval`, and date range:

```
curl --location 'https://api.upstox.com/v3/historical-
candle/NSE_EQ%7CINE002A01018/days/1/2025-03-04/2025-01-01' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

Historical Candle Data V3 supports `minutes`, `hours`, `days`, `weeks`, and `months` units, with daily data available from January 2000 and intraday from January 2022. Review the availability table on the Historical Candle Data V3 page for the exact limits per unit.

Backtest F&O (expired contracts)

Derivatives contracts expire, so a realistic F&O backtest needs the exact contract that traded during your test window. The flow adds two steps to discover expired contracts before fetching candles.

Step 1 — Resolve the underlying instrument key

Find the `instrument_key` for the underlying index or stock. For an index:

```
curl --location 'https://api.upstox.com/v2/instruments/search?
query=NIFTY&exchanges=NSE&segments=INDEX&records=5' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

This returns `NSE_INDEX|Nifty 50` for the Nifty 50 index. For stock derivatives, search the equity segment (as in the cash example) to get the underlying key such as `NSE_EQ|INE002A01018`.

Step 2 — List available expiries

Pass the underlying key to Get Expiries to retrieve every past expiry date for that underlying:

```
curl --location 'https://api.upstox.com/v2/expired-instruments/expiries?
instrument_key=NSE_INDEX%7CNifty%2050' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

The response is a list of dates in `YYYY-MM-DD` format. Pick the expiry that covers your backtest window.

Step 3 — Get the expired contracts

Use the underlying key and a chosen `expiry_date` to list the contracts that settled on that date. For options, call Get Expired Option Contracts:

```
curl --location 'https://api.upstox.com/v2/expired-instruments/option/contract?
instrument_key=NSE_INDEX%7CNifty%2050&expiry_date=2025-04-24' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

For futures, call Get Expired Future Contracts with the same parameters. Each item in the response includes an `expired_instrument_key` (for example, `NSE_FO|53806|24-04-2025`), along with the strike price, option type, and trading symbol. Select the contract you want to test and copy its `expired_instrument_key`.

Step 4 — Fetch expired historical candles

Pass the `expired_instrument_key` (URL-encoded), an `interval`, and a date range to Get Expired Historical Candle Data:

```
curl --location 'https://api.upstox.com/v2/expired-instruments/historical-candle/NSE_FO%7C53806%7C24-04-2025/day/2025-04-24/2025-03-24' \
--header 'Accept: application/json' \
--header 'Authorization: Bearer {your_access_token}'
```

Each candle is an array of `[timestamp, open, high, low, close, volume, open_interest]`. Open interest is populated for derivatives, which lets you factor liquidity and positioning into your F&O backtests. See Get Expired Historical Candle Data for the supported intervals and per-interval history limits.

Response format

All candle endpoints return the same OHLC array structure, so your backtesting engine can parse active and expired data with one code path:

```
{
  "status": "success",
  "data": {
    "candles": [
      [
        "2025-04-24T00:00:00+05:30",
        125.35,
        126.8,
        122.1,
        123.45,
        1542678,
        184632
      ]
    ]
  }
}
```

Index	Field	Description
0	Timestamp	Start time of the candle.
1	Open	Opening price for the timeframe.
2	High	Highest traded price.
3	Low	Lowest traded price.
4	Close	Closing price for the timeframe.

Index	Field	Description
5	Volume	Total quantity traded.
6	Open Interest	Outstanding derivative contracts (`0` for cash instruments).

Best practices

- **Resolve keys once, then cache them.** Instrument and expiry lists change slowly. Store the keys you resolve so a long backtest does not re-query them on every run.
- **Respect interval history limits.** Finer intervals (such as `1minute`) cover shorter windows than daily or weekly candles. Split long backtests into chunks that fit each interval's limit, and page through date ranges.
- **Handle rate limits.** Backtests can generate many requests. Follow the rate limiting guidelines and add retry/backoff logic. A